

Byung Gook Park

professor

基本信息

姓名: Byung Gook Park

所属院校: 首尔国立大学

地区: South Korea

年龄范围: 50岁以上

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

存储器设计与架构

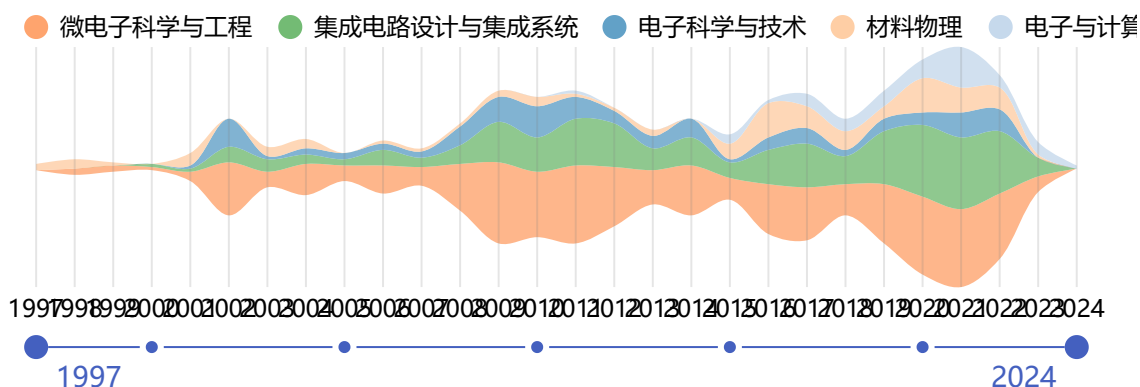
个人简介

Byung Gook Park is a Professor of Electrical Engineering at Seoul National University. His research focuses on semiconductor devices, nanoelectronics, optical devices, and VLSI technology. He has published extensively in top-tier journals and holds several patents in the field of semiconductor device fabrication and optimization. His notable contributions include advancements in tunneling field-effect transistors (TFETs), three-dimensional NAND flash architecture, and analog synaptic behavior of silicon nitride memristors.

工作经历

Seoul National University - Professor

近年发表的论文



A vision sensor for simultaneous imaging and distance sensing in real-time

JOURNAL OF MICROMECHANICS AND MICROENGINEERING

Haan, Keewoong; Chun, Kukjin; Park, Byung-Gook; Kim, Hyeon Cheol; Kim, Bonghwan

被引用次数: 0

2024

Negative capacitance field-effect transistor with hetero-dielectric structure for suppression of reverse drain induced barrier lowering

SOLID-STATE ELECTRONICS

Kim, Dong-Oh; Lee, Kitae; Kim, Changha; Kim, Sihyun; Kim, Hyun-Min; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 1

2023

Effect of layer-specific synaptic retention characteristics on the accuracy of deep neural networks

SOLID-STATE ELECTRONICS

Yoo, Ho-Nam; Park, Min-Kyu; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 0

2023

Synaptic array using multi-level AND flash memory cells for hardware-based neural networks

SOLID-STATE ELECTRONICS

Lee, Soochang; Oh, Seongbin; Woo, Sung Yun; Park, Byung-Gook; Lee, Jong -Ho

被引用次数: 0

2023

Electric-Field-Induced Metal Filament Formation in Cobalt-Based CBRAM Observed by TEM

ACS APPLIED ELECTRONIC MATERIALS

Choi, Yeon-Joon; Bang, Suhyun; Kim, Tae-Hyeon; Hong, Kyungho; Kim, Sungjoon; Kim, Sungjun; Park, Byung-Gook; Choi, Woo Young

被引用次数: 8

2023

Efficient Hybrid Training Method for Neuromorphic Hardware Using Analog Nonvolatile Memory

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Kwon, Dongseok; Woo, Sung Yun; Hwang, Joon; Kim, Hyeongsu; Bae, Jong-Ho; Shin, Wonjun; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 0

2023

Data-driven multi-objective optimization with neural network-based sensitivity analysis for semiconductor devices

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

Oh, Min-Hye; Lee, Kitae; Kim, Sihyun; Park, Byung-Gook

被引用次数: 2

2023

Optimization of the structural complexity of artificial neural network for hardware-driven neuromorphic computing application

APPLIED INTELLIGENCE

Mohanan, Kannan Udaya; Cho, Seongjae; Park, Byung-Gook

被引用次数: 13

2023

A novel pathway to construct gas concentration prediction model in real-world applications: Data augmentation; fast prediction; and interpolation and extrapolation

SENSORS AND ACTUATORS B-CHEMICAL

Kim, Jaehyeon; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Choi, Woo Young; Kim, Jae -Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 4

2023

Hardware-Based Noisy Deep Q-Networks Using Low-Frequency Noise of Synaptic Devices for Efficient Exploration

IEEE ELECTRON DEVICE LETTERS

Kim, Jangsaeng; Seo, Young-Tak; Shin, Wonjun; Choi, Woo Young; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 5

2023

Fuse Devices for Pruning in Memristive Neural Network

IEEE ELECTRON DEVICE LETTERS

Kim, Tae-Hyeon; Hong, Kyungho; Kim, Sungjoon; Park, Jinwoo; Youn, Sangwook; Lee, Jong-Ho; Park, Byung-Gook; Kim, Hyungjin; Choi, Woo Young

被引用次数: 5

2023

Demonstration of Pulse Width Modulation Function Using Single Positive Feedback Device for Neuron

IEEE ELECTRON DEVICE LETTERS

Woo, Sung Yun; Kwon, Dongseok; Park, Byung-Gook; Lee, Jong-Ho; Bae, Jong-Ho

被引用次数: 2

2023

Floating Fin Shaped Stacked Nanosheet MOSFET for Low Power Logic Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kim, Munhyeon; Kim, Sihyun; Lee, Kitae; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 1

2023

Impact of interlayer insulator formation methods on HfO_x ferroelectricity in the metal-ferroelectric-insulator-semiconductor stack

APPLIED PHYSICS LETTERS

Min, Kyung Kyu; Kwon, Seok Jin; Kim, Yeonwoo; Yu, Junsu; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 5

2022

An integrate-and-fire neuron with capacitive trans-impedance amplifier for improving linearity in Spiking Neural Networks

SOLID-STATE ELECTRONICS

Cha, Youngsan; Park, Kyungchul; Lee, Jong-Ho; Park, Byung-Gook

被引用次数: 6

2022

Impact of the leakage current of an AND-type synapse array on spiking neural networks

SOLID-STATE ELECTRONICS

Jeon, Bosung; Song, Seunghwan; Hwang, Sungmin; Jang, Taejin; Park, Kyungchul; Park, Jonghyuk; Lee, Jong-Ho; Park, Byung-Gook

被引用次数: 1

2022

Effect of weight overlap region on neuromorphic system with memristive synaptic devices

CHAOS SOLITONS & FRACTALS

Lee, Geun Ho; Kim, Tae-Hyeon; Song, Min Suk; Park, Jinwoo; Kim, Sungjoon; Hong, Kyungho; Kim, Yoon; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 13

2022

Synergistic improvement of sensing performance in ferroelectric transistor gas sensors using remnant polarization

MATERIALS HORIZONS

Shin, Wonjun; Yim, Jiyong; Bae, Jong-Ho; Lee, Jung-Kyu; Hong, Seongbin; Kim, Jaehyeon; Jeong, Yujeong; Kwon, Dongseok; Koo, Ryun-Han; Jung, Gyuweon; Han, Changhyeon; Kim, Jeonghan; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 19

2022

Comprehensive and accurate analysis of the working principle in ferroelectric tunnel junctions using low-frequency noise spectroscopy

NANOSCALE

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Yim, Jiyong; Kwon, Dongseok; Kim, Yeonwoo; Yu, Junsu; Hwang, Joon; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 29

2022

Ferroelectricity of pure HfO_x in metal-ferroelectric-insulator-semiconductor stacks and its memory application

APPLIED SURFACE SCIENCE

Min, Kyung Kyu; Yu, Junsu; Kim, Yeonwoo; Kim, Chae Soo; Jang, Taejin; Hwang, Sungmin; Kim, Hyungjin; Lee, Jong-Ho; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 18

2022

Medium-Temperature-Oxidized GeO_x Resistive-Switching Random-Access Memory and Its Applicability in Processing-in-Memory Computing

NANOSCALE RESEARCH LETTERS

Mohanan, Kannan Udaya; Cho, Seongjae; Park, Byung-Gook

被引用次数: 12

2022

Highly-packed self-assembled graphene oxide film-integrated resistive random-access memory on a silicon substrate for neuromorphic application

NANOTECHNOLOGY

Choi, Hyun-Seok; Lee, Jihye; Kim, Boram; Lee, Jaehong; Park, Byung-Gook; Kim, Yoon; Hong, Suck Won

被引用次数: 5

2022

4-bit Multilevel Operation in Overshoot Suppressed Al₂O₃/TiO_x Resistive Random-Access Memory Crossbar Array

ADVANCED INTELLIGENT SYSTEMS

Kim, Sungjoon; Park, Jinwoo; Kim, Tae-Hyeon; Hong, Kyungho; Hwang, Yeongjin; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 44

2022

A Fast Weight Transfer Method for Real-Time Online Learning in RRAM-Based Neuromorphic System

IEEE ACCESS

Kim, Min-Hwi; Lee, Sin-Hyung; Kim, Sungjun; Park, Byung-Gook

被引用次数: 9

2022

On-Chip Trainable Spiking Neural Networks Using Time-To-First-Spike Encoding

IEEE ACCESS

Im, Jiseong; Kim, Jaehyeon; Yoo, Ho-Nam; Baek, Jong-Won; Kwon, Dongseok; Oh, Seongbin; Kim, Jangsaeng; Hwang, Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 2

2022

Fully integrated FET-type gas sensor with optimized signal-to-noise ratio for H₂S gas detection

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 17

2022

Optimization of channel structure and bias condition for signal-to-noise ratio improvement in Si-based FET-type gas sensor with horizontal floating-gate

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 17

2022

Spiking Neural Network With Weight-Sharing Synaptic Array for Multi-input Processing

IEEE ELECTRON DEVICE LETTERS

Song, Seunghwan; Kim, Munhyeon; Jeon, Bosung; Ryu, Donghyun; Kim, Sihyun; Lee, Kitae; Lee, Jong-Ho; Kim, Jae-Joon; Shim, Wonbo; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 3

2022

Highly Efficient Self-Curing Method in MOSFET Using Parasitic Bipolar Junction Transistor

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Kwon, Dongseok; Hwang, Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 8

2022

Investigation of Low-Frequency Noise Characteristics of Ferroelectric Tunnel Junction: From Conduction Mechanism and Scaling Perspectives

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kwon, Dongseok; Koo, Ryun-Han; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 17

2022

Damage-Induced Ferroelectricity in HfO_x-Based Thin Film

IEEE ELECTRON DEVICE LETTERS

Min, Kyung Kyu; Kim, Hyun-Min; Kim, Yeonwoo; Kim, Changha; Yu, Junsu; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 3

2022

Capacitor-Based Synaptic Devices for Hardware Spiking Neural Networks

IEEE ELECTRON DEVICE LETTERS

Hwang, Sungmin; Yu, Junsu; Lee, Geun Ho; Song, Min Suk; Chang, Jeessoo; Min, Kyung Kyu; Jang, Taejin; Lee, Jong-Ho; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 29

2022

Effects of Channel Length Scaling on the Signal-to-Noise Ratio in FET-Type Gas Sensor With Horizontal Floating-Gate

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 20

2022

Variation-Tolerant Capacitive Array for Binarized Neural Network

IEEE ELECTRON DEVICE LETTERS

Kim, Hyeongsu; Woo, Sung Yun; Lee, Soochang; Seo, Young-Tak; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 3

2022

Effects of High-Pressure Annealing on the Low-Frequency Noise Characteristics in Ferroelectric FET

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Sihyun; Lee, Kitae; Kwon, Dongseok; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 40

2022

SiO₂ Fin-Based Flash Synaptic Cells in AND Array Architecture for Binary Neural Networks

IEEE ELECTRON DEVICE LETTERS

Lee, Soochang; Kim, Hyeongsu; Lee, Sung-Tae; Park, Byung-Gook; Lee, Jong-Ho

Incorporating Bottom-Up Approach Into Device/Circuit Co-Design for SRAM-Based Cache Memory Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tayal, Shubham; Smaani, Billel; Rahi, Shiromani Balmukund; Upadhyay, Abhishek Kumar; Bhattacharya, Sandip; Ajayan, J.; Jena, Biswajit; Myeong, Ilho; Park, Byung-Gook; Song, Young Suh

被引用次数: 12

2022

Highly Linear Analog Spike Processing Block Integrated With an AND-Type Flash Array and CMOS Neuron Circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kyu-Ho; Kwon, Dongseok; Woo, Sung Yun; Ko, Jong Hyun; Choi, Woo Young; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 7

2022

Effect of Program Error in Memristive Neural Network With Weight Quantization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Tae-Hyeon; Kim, Sungjoon; Hong, Kyungho; Park, Jinwoo; Youn, Sangwook; Lee, Jong-Ho; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 24

2022

Effects of Postdeposition Annealing Ambience on NO₂ Gas Sensing Performance in Si-Based FET-Type Gas Sensor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 10

2022

Novel Dual Liner Process for Side-Shielded Forksheet Device With Superior Design Margin

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Munhyeon; Lee, Kitae; Kim, Sihyun; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 0

2022

Investigation of Device Performance for Fin Angle Optimization in FinFET and Gate-All-Around FETs for 3 nm-Node and Beyond

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Soyoun; Lee, Kitae; Kim, Sihyun; Kim, Munhyeon; Lee, Jong-Ho; Kim, Sangwan; Park, Byung-Gook

被引用次数: 18

2022

Comprehensive TCAD-Based Validation of Interface Trap-Assisted Ferroelectric Polarization in Ferroelectric-Gate Field-Effect Transistor Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kitae; Kim, Sihyun; Kim, Munhyeon; Lee, Jong-Ho; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 13

2022

Ferroelectric-Metal Field-Effect Transistor With Recessed Channel for 1T-DRAM Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Lee, Kitae; Kim, Sihyun; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 5

2022

Investigation on Variability of Ferroelectric-Gate Field-Effect Transistor Memory by Random Spatial Distribution of Interface Trap

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Lee, Kitae; Kim, Sihyun; Kim, Munhyeon; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 2

2022

Analog synaptic devices applied to spiking neural networks for reinforcement learning applications

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Jangsaeng; Lee, Soochang; Kim, Chul-Heung; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 2

2022

Lamination method for improved polarization-leakage current relation in HfO₂-based metal/ferroelectric/insulator/semiconductor structure

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Yeonwoo; Min, Kyung Kyu; Yu, Junsu; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 10

2022

Enhanced Current-voltage Nonlinearity by Controlling Oxygen Concentration of TiO_x Buffer Layer for RRAM Passive Crossbar Array

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Tae-Hyeon; Kim, Sungjoon; Hong, Kyungho; Kim, Hyungjin; Park, Byung-Gook

被引用次数: 4

2022

Highly Reliable Physical Unclonable Functions using Memristor Crossbar with Tunneling Conduction

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Park, Jinwoo; Kim, Tae-Hyeon; Kim, Sungjoon; Song, Min Suk; Youn, Sangwook; Hong, Kyungho; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 3

2022

On-chip adaptive matching learning with charge-trap synapse device and ReLU activation circuit

SOLID-STATE ELECTRONICS

Ahn, Ji-Hoon; Choi, Hyun-Seok; Kim, Jung Nam; Park, Byung-Gook; Kim, Sungjun; Lee, Jaehong; Kim, Yoon

被引用次数: 12

2021

Improvement of self-heating effect in Ge vertically stacked GAA nanowire pMOSFET by utilizing Al₂O₃ for high-performance logic device and electrical/thermal co-design

JAPANESE JOURNAL OF APPLIED PHYSICS

Song, Young Suh; Kim, Sangwan; Kim, Garam; Kim, Hyunwoo; Lee, Jong-Ho; Kim, Jang Hyun; Park, Byung-Gook

Nanoscale wedge resistive-switching synaptic device and experimental verification of vector-matrix multiplication for hardware neuromorphic application

JAPANESE JOURNAL OF APPLIED PHYSICS

Kim, Min-Hwi; Cho, Seongjae; Park, Byung-Gook

被引用次数: 13

2021

Multilevel switching memristor by compliance current adjustment for off-chip training of neuromorphic system

CHAOS SOLITONS & FRACTALS

Kim, Tae-Hyeon; Kim, Sungjoon; Hong, Kyungho; Park, Jinwoo; Hwang, Yeongjin; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 41

2021

Effect of charge storage engineering on the NO₂ gas sensing properties of a WO₃ FET-type gas sensor with a horizontal floating-gate

NANOSCALE

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Lee, Chayoung; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 18

2021

Interlayer engineering for enhanced ferroelectric tunnel junction operations in HfO_x-based metal-ferroelectric-insulator-semiconductor stack

NANOTECHNOLOGY

Min, Kyung Kyu; Yu, Junsu; Kim, Yeonwoo; Lee, Jong-Ho; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 12

2021

A novel physical unclonable function (PUF) using 16 x 16 pure-HfO_x ferroelectric tunnel junction array for security applications

NANOTECHNOLOGY

Yu, Junsu; Min, Kyung Kyu; Kim, Yeonwoo; Kim, Sihyun; Hwang, Sungmin; Kim, Tae-Hyeon; Kim, Changha; Kim, Hyungjin; Lee, Jong-Ho; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 9

2021

3-bit multilevel operation with accurate programming scheme in TiO_x/Al₂O₃ memristor crossbar array for quantized neuromorphic system

NANOTECHNOLOGY

Kim, Tae-Hyeon; Lee, Jaewoong; Kim, Sungjoon; Park, Jinwoo; Park, Byung-Gook; Kim, Hyungjin

被引用次数: 30

2021

Hardware -based spiking neural network architecture using simplified backpropagation algorithm and homeostasis functionality

NEUROCOMPUTING

Kim, Jangsaeng; Kwon, Dongseok; Woo, Sung Yun; Kang, Won-Mook; Lee, Soochang; Oh, Seongbin; Kim, Chul-Heung; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 19

2021

On-chip trainable hardware-based deep Q-networks approximating a backpropagation algorithm

NEURAL COMPUTING & APPLICATIONS

Kim, Jangsaeng; Kwon, Dongseok; Woo, Sung Yun; Kang, Won-Mook; Lee, Soochang; Oh, Seongbin; Kim, Chul-Heung; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 8

2021

Direct Gradient Calculation: Simple and Variation-Tolerant On-Chip Training Method for Neural Networks

ADVANCED INTELLIGENT SYSTEMS

Kim, Hyungyo; Hwang, Joon; Kwon, Dongseok; Kim, Jangsaeng; Park, Min-Kyu; Im, Jiseong; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 6

2021

Low-Power Adaptive Integrate-and-Fire Neuron Circuit Using Positive Feedback FET Co-Integrated With CMOS

IEEE ACCESS

Kwon, Min-Woo; Park, Kyungchul; Park, Byung-Gook

被引用次数: 6

2021

Integrate-and-Fire Neuron Circuit With Synaptic Off-Current Blocking Operation

IEEE ACCESS

Song, Seunghwan; Jeon, Bosung; Hwang, Sungmin; Baek, Myung-Hyun; Lee, Jong-Ho; Park, Byung-Gook

被引用次数: 1

2021

Selected Bit-Line Current PUF: Implementation of Hardware Security Primitive Based on a Memristor Crossbar Array

IEEE ACCESS

Kim, Dayoung; Kim, Tae-Hyeon; Choi, Yunyeong; Lee, Geun Ho; Lee, Jungwon; Sun, Wookyoung; Park, Byung-Gook; Kim, Hyungjin; Shin, Hyungsoon

被引用次数: 4

2021

Hardware-Based Spiking Neural Network Using a TFT-Type AND Flash Memory Array Architecture Based on Direct Feedback Alignment

IEEE ACCESS

Kang, Won-Mook; Kwon, Dongseok; Woo, Sung Yun; Lee, Soochang; Yoo, Honam; Kim, Jangsaeng; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 12

2021

Spiking Neural Networks With Time-to-First-Spike Coding Using TFT-Type Synaptic Device Model

IEEE ACCESS

Oh, Seongbin; Lee, Soochang; Woo, Sung Yun; Kwon, Dongseok; Im, Jiseong; Hwang, Joon; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 5

2021

Efficient fusion of spiking neural networks and FET-type gas sensors for a fast and reliable artificial olfactory system

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Kim, Jaehyeon; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 26

2021

Low-power and reliable gas sensing system based on recurrent neural networks

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 19

2021

Comparison of the characteristics of semiconductor gas sensors with different transducers fabricated on the same substrate

SENSORS AND ACTUATORS B-CHEMICAL

Jung, Gyuweon; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 35

2021

Improved signal-to-noise-ratio of FET-type gas sensors using body bias control and embedded micro-heater

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Hong, Seongbin; Jung, Gyuweon; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Jang, Dongkyu; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 22

2021

Analytically and empirically consistent characterization of the resistive switching mechanism in a Ag conducting-bridge random-access memory device through a pseudo-liquid interpretation approach

PHYSICAL CHEMISTRY CHEMICAL PHYSICS

Choi, Yeon-Joon; Bang, Suhyun; Kim, Tae-Hyeon; Hong, Kyungho; Kim, Sungjoon; Kim, Sungjun; Cho, Seongjae; Park, Byung-Gook

被引用次数: 1

2021

Suppression of Statistical Variability in Stacked Nanosheet Using Floating Fin Structure

IEEE ELECTRON DEVICE LETTERS

Kim, Munhyeon; Kim, Sihyun; Lee, Kitae; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 1

2021

Double-Gated Ferroelectric-Gate Field-Effect-Transistor for Processing in Memory

IEEE ELECTRON DEVICE LETTERS

Kim, Munhyeon; Lee, Kitae; Kim, Sihyun; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 11

2021

Effect of Lateral Charge Diffusion on Retention Characteristics of 3D NAND Flash Cells

IEEE ELECTRON DEVICE LETTERS

Yoo, Ho-Nam; Choi, Bongsik; Back, Jong-Won; Kang, Ho-Jung; Kwon, Eunmee; Chung, Sungyong; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 13

2021

Vertically-Stacked Si_{0.2}Ge_{0.8} Nanosheet Tunnel FET With 70 mV/Dec Average Subthreshold Swing

IEEE ELECTRON DEVICE LETTERS

Lee, Ryoongbin; Lee, Junil; Lee, Kitae; Kim, Soyoun; Ahn, Hyunho; Kim, Sihyun; Kim, Hyun-Min; Kim, Changha; Lee, Jong-Ho; Kim, Sangwan; Park, Byung-Gook

被引用次数: 9

2021

Physical Unclonable Functions Using Ferroelectric Tunnel Junctions

IEEE ELECTRON DEVICE LETTERS

Kim, Sihyun; Lee, Kitae; Oh, Min-Hye; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 8

2021

Impacts of Program/Erase Cycling on the Low-Frequency Noise Characteristics of Reconfigurable Gated Schottky Diodes

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kwon, Dongseok; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 10

2021

Effects of Process-Induced Defects on Polarization Switching in Ferroelectric Tunneling Junction Memory

IEEE ELECTRON DEVICE LETTERS

Lee, Kitae; Kim, Sihyun; Lee, Jong-Ho; Park, Byung-Gook; Kwon, Daewoong

被引用次数: 19

2021

Investigation of Low-Frequency Noise Characteristics in Gated Schottky Diodes

IEEE ELECTRON DEVICE LETTERS

Kwon, Dongseok; Shin, Wonjun; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 12

2021

Trap-Induced Data-Retention-Time Degradation of DRAM and Improvement Using Dual Work-Function Metal Gate

IEEE ELECTRON DEVICE LETTERS

Kim, Kyoung Yeon; Min, Kyung Kyu; Park, Byung-Gook

被引用次数: 6

2021

Effect of Random Dopant Fluctuation on Data Retention Time Distribution in DRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Kyoung Yeon; Park, Byung-Gook

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IEEE ELECTRON DEVICE LETTERS

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IEICE TRANSACTIONS ON ELECTRONICS

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IEEE TRANSACTIONS ON NANOTECHNOLOGY

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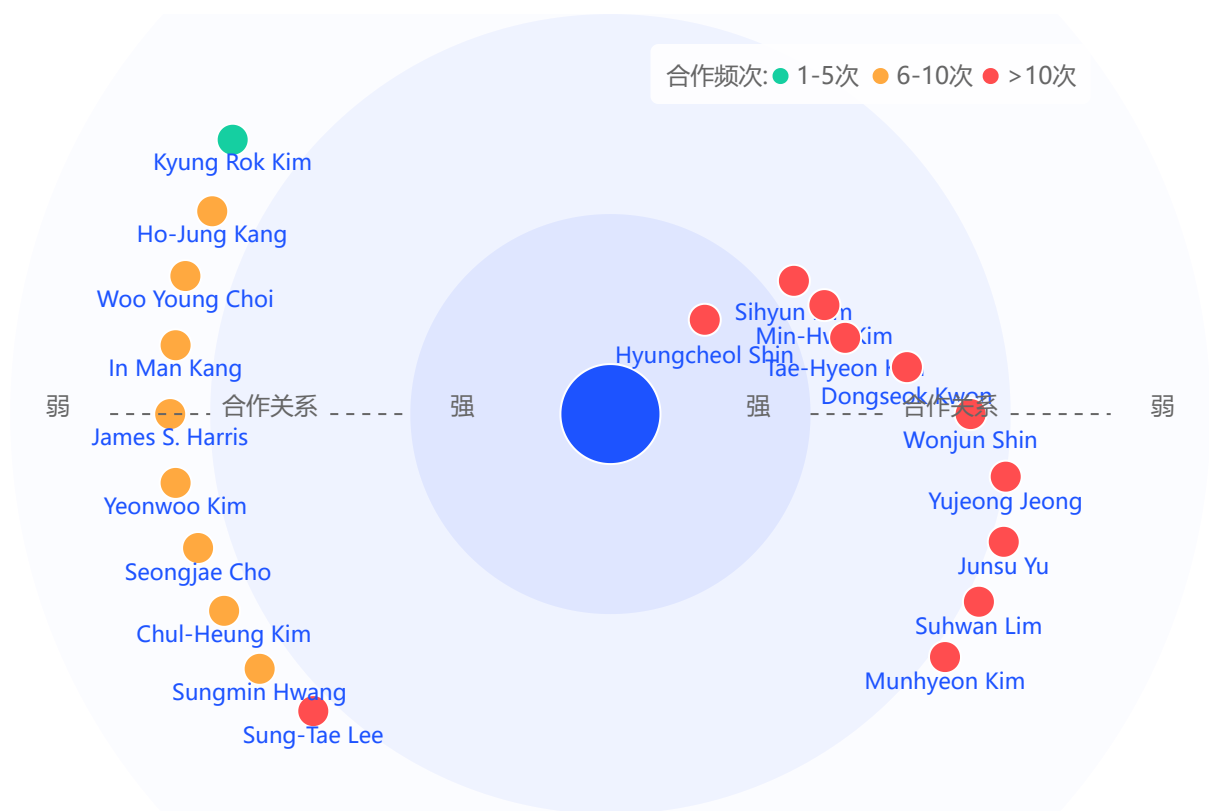
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